

GRAIL

NASDAQ : GRAL ·
Young-Company DCF
(Damodaran)

\$3.1B mkt cap · 44M sh · \$823M cash, zero debt · \$0.823B cash, no debt; ~43M shares (low float); Illumina holds ~12.5%;
Revenue-generating: \$147M FY25 (+17%), Galleri self-pay; burn ~\$300M/yr (~2.5yr); NHS-Galleri missed primary endpoint
(Feb'26); FDA PMA filed; Medicare law signed

\$73.00

THE CENTRAL QUESTION

Does ~\$3B / ~20x sales price GRAIL's multi-gate regulatory option fairly — after the NHS-Galleri miss, but with the FDA filing and Medicare law in hand?

GRAIL trades at ~\$3B (~\$73/sh on just ~43M shares) on ~\$147M FY25 revenue and ~\$823M cash. Galleri — a blood test for 50+ cancers, self-pay today — is a multi-gate regulatory option: FDA PMA (filed Jan 2026) → Medicare coverage (signed Feb 2026; ≥2028) → broad adoption. The shape is unusual: the pivotal NHS-Galleri trial MISSED its endpoint (-50% stock), but the FDA filing rests on the positive PATHFINDER 2 and the Medicare law passed — partly broken, partly de-risked. The finding: at \$73 the market roughly prices the modal regulatory-success case, with FDA-rejection downside and broad-adoption upside about balanced.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$79.54
expected fair value today
+9.0% vs spot \$73.00

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$123.72	\$192.59	\$300.05	\$467.89
1.69× spot	2.64× spot	4.11× spot	6.41× spot

WEIGHTING RATIONALE

Ultra Bear 20% FDA rejects; no coverage; near-total loss.

Bear 28% FDA late; Medicare narrow; sub-scale; low teens.

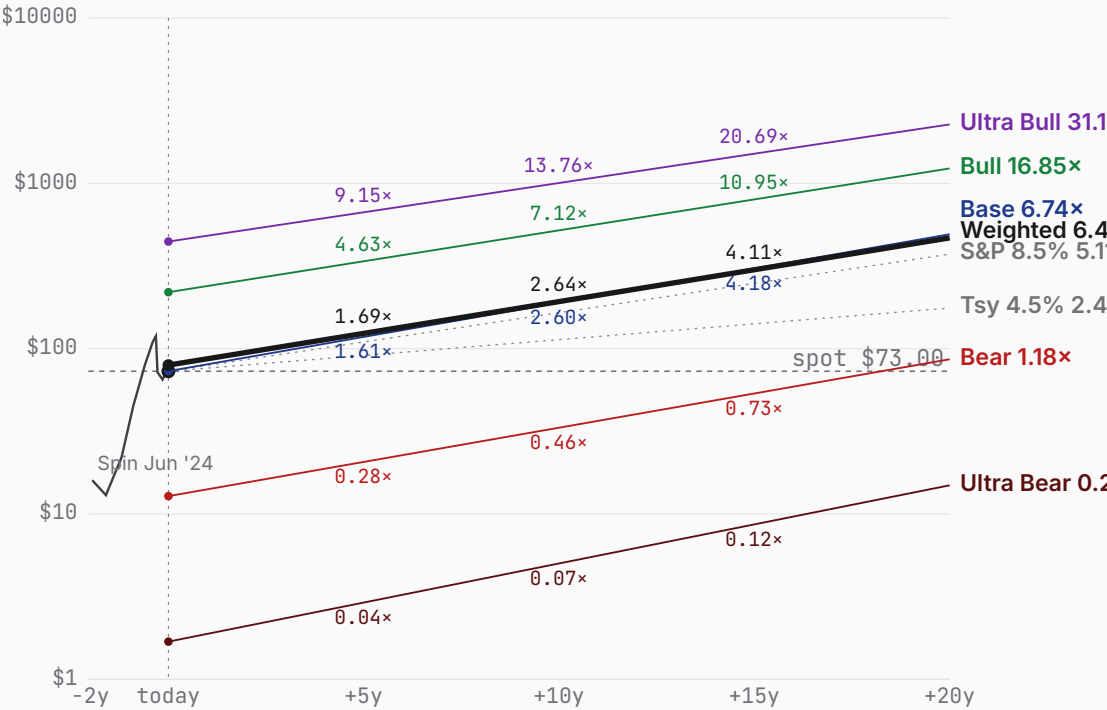
Base 34% FDA 2027 + Medicare 2028; leader; ≈ spot.

Bull 13% Broad adoption; category leader; ~3×.

Ultra Bull 5% MCED standard of care; dominant; ~6×.

Bull (13%) + Ultra Bull (5%) together contribute \$50.75 — 64% of the \$79.54 expected value despite 18% combined probability. Today's spot (\$73.00) sits 8% below the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED



ULTRA BEAR	\$1.69	BEAR	\$12.82	BASE	\$73.13	BULL	\$219.48	ULTRA BULL	\$444.35
	-97.7% vs spot		-82.4% vs spot		+0.2% vs spot		+200.7% vs spot		+508.7% vs spot
TAM 1.3% P(fail) 40% S2C 1.0 Prob 20%		TAM 4.1% P(fail) 22% S2C 1.1 Prob 28%		TAM 10.4% P(fail) 12% S2C 1.3 Prob 34%		TAM 18.0% P(fail) 6% S2C 1.5 Prob 13%		TAM 27.2% P(fail) 3% S2C 1.7 Prob 5%	
FDA rejects; no coverage; self-pay niche.		FDA late; Medicare narrow; modest volume.		FDA 2027 + Medicare 2028; category leader.		Broad adoption; annual-repeat screening.		MCED standard of care; GRAIL dominant.	

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LONG HORIZONS · STRUCTURAL SHIFTS · IMAGINATION

GRAIL scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 20% Bear 28% Base 34% Bull 13% Ultra Bull 5% · Weighted expected \$79.54 (+9.0% vs spot)

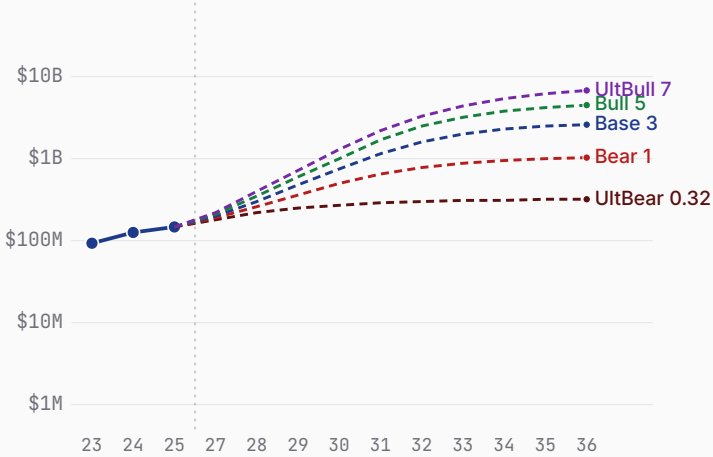
ULTRA BEAR	\$1.69	BEAR	\$12.82	BASE	\$73.13	BULL	\$219.48	ULTRA BULL	\$444.35
	-97.7% vs spot		-82.4% vs spot		+0.2% vs spot		+200.7% vs spot		+508.7% vs spot
Probability 20%		Probability 28%		Probability 34%		Probability 13%		Probability 5%	
FDA rejects; no coverage; self-pay niche.		FDA late; Medicare narrow; modest volume.		FDA 2027 + Medicare 2028; category leader.		Broad adoption; annual-repeat screening.		MCED standard of care; GRAIL dominant.	
WHY 20%		WHY 28%		WHY 34%		WHY 13%		WHY 5%	
The NHS-Galleri miss is a real, fired datapoint — the only RCT of any MCED failed its stage-shift primary endpoint, hardening FDA and payer skepticism. 40% within-scenario failure reflects genuine PMA-rejection risk. 20% scenario weight.		FDA approves but adoption disappoints — Medicare narrow, private payers await mortality proof, ASP compresses. 22% failure probability. 28% weight as a very plausible 'approved but sub-scale' outcome.		Requires FDA approval (on PATHFINDER 2, which was positive) + Medicare activation (the law is already signed) + broad coverage — the machinery is in motion. 12% failure probability. 34% weight as the central, modal outcome.		A chain: clean FDA + Medicare + broad private coverage + annual-repeat adoption + category leadership. Each plausible at 50-70%; jointly ~13%. GRAIL's breadth and only-RCT dataset are the edge.		Tail of tails: MCED standard of care + eventual mortality proof + GRAIL dominance + premium terminal. Individually 40-60%; jointly ~5%. The asymmetric upside the ~43M share count amplifies.	
WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS		WHAT HAPPENS	
The FDA PMA is rejected or stalled — the NHS-Galleri primary-endpoint miss (no significant stage III+IV reduction) hardens FDA/USPSTF skepticism about clinical utility, and Galleri stays a self-pay LDT niche. Test volume plateaus, the Medicare benefit never activates (it is gated on FDA approval), and GRAIL burns its ~\$823M cash defending the franchise.		The FDA approves but late, and Medicare coverage arrives narrow and slow; private payers wait for the mortality data the NHS trial did not deliver. Galleri reaches modest reimbursed volume (~\$1B revenue by 2035) at thin-but-improving margins — a real test, sub-scale economics.		The modal path: the FDA approves Galleri (~2027) on the PATHFINDER 2 evidence, Medicare coverage activates (~2028), and broad private coverage follows — Galleri becomes the leading reimbursed MCED test at ~\$2.6B revenue by 2035 (~10% of a ~\$25B screening market) at ~30% operating margins.		The bull: FDA + Medicare clear cleanly, broad private coverage and annual-repeat screening take hold, and GRAIL extends its first-mover, broadest-test, only-RCT-dataset lead — ~\$4.5B revenue by 2035 at ~36% operating margins. The screening category compounds and GRAIL leads it.		The tail: MCED becomes standard of care, the extended NHS follow-up eventually shows a mortality benefit, and GRAIL is the dominant platform — ~\$6.8B revenue by 2035 at ~40% margins with a premium terminal.	
With a 40% within-scenario failure probability and revenue stalled near \$0.3B, the equity collapses toward the cash-and-distress floor — a near-total loss from today's \$73.		The DCF lands in the low teens — well below spot. This is the 'approved, but adoption disappoints' world the NHS miss makes plausible.		DCF ~\$82/share — roughly today's price. The market is pricing this modal regulatory-success case; the small ~50M share count makes the per-share value swing hard on each gate.		DCF ~\$232/share — ~3× today's price. This is the scenario the regulatory machinery (PMA filed + Medicare law) keeps alive despite the NHS miss.		DCF ~\$457/share — ~6× spot. The asymmetric upside the low share count amplifies; individually each gate is plausible, jointly ~5%.	

GRAIL business snapshot

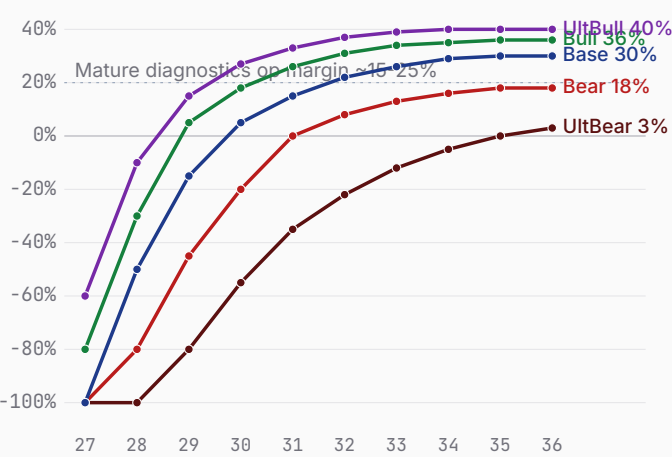
Bear \$12.82 Base \$73.13 Bull \$219.48

FY26-FY35 scenario projections (revenue-generating; pre-profit) · fiscal years end Dec 31 · Q1 2026 10-Q, FY25 results

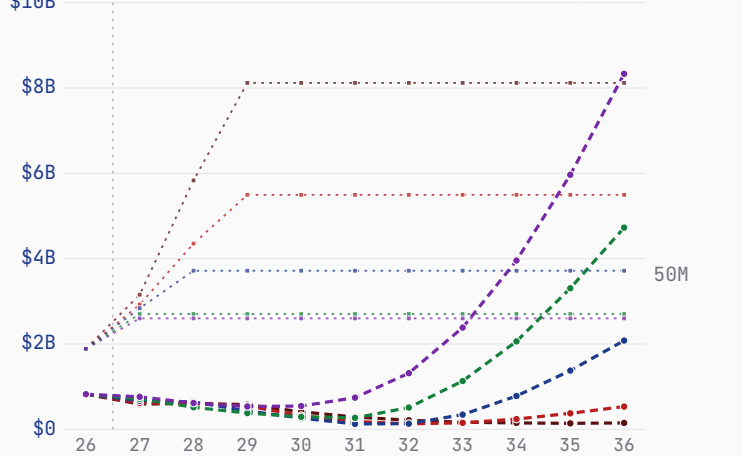
Revenue — history + scenarios to FY36 (log)



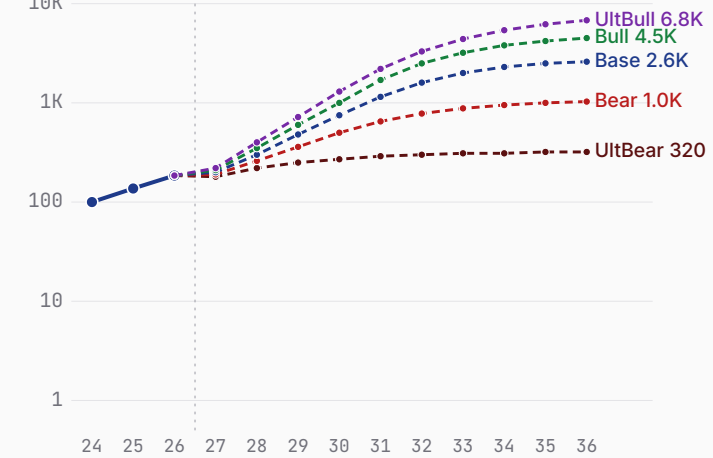
Path to profitability — op margin (FY27→FY36)



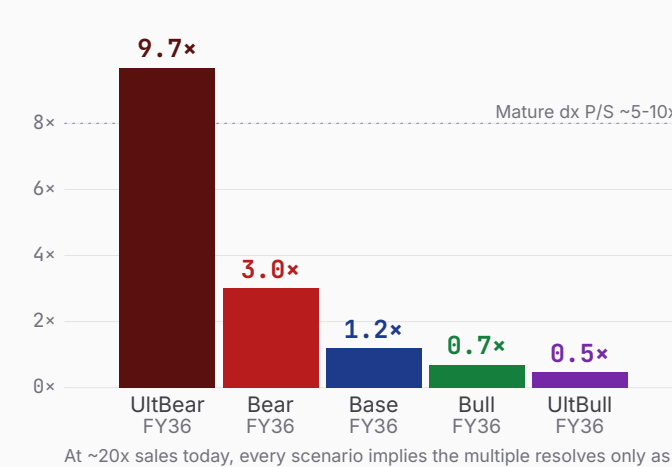
Cash + dilution (FY26 → FY36)



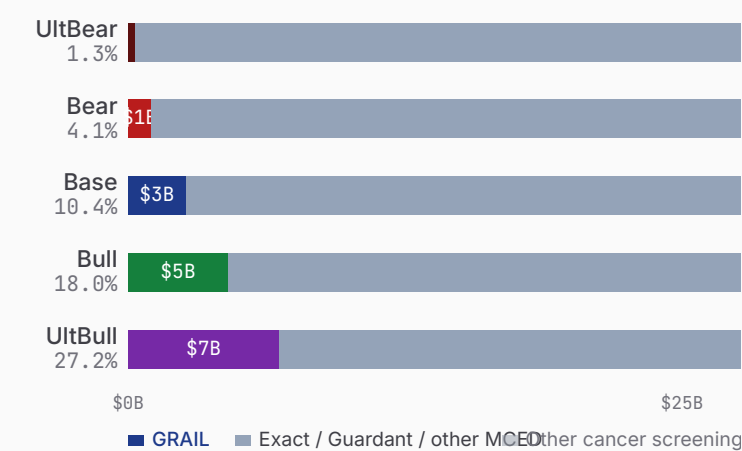
Galleri tests sold (000s/yr, log)



\$3.10B mkt cap as P/S on FY36 rev



\$25B FY35 MCED screening TAM — GRAIL share



Sources: GRAIL Q1 2026 10-Q (cash \$0.823B), FY2025 results (\$147M revenue, net loss), NHS-Galleri (primary endpoint missed Feb'26) + PATHFINDER 2 (positive) readouts, FDA PMA filing, Medicare MCED Coverage Act. Revenue modeled as reimbursed test volume x ASP; FCF NOL-shielded pre-tax over the window (documented simplification). TAM: MCED screening serviceable slice, ~\$25B annual by FY35.

GRAIL 7 Powers · the moat, scored

Bear \$12.82 Base \$73.13 Bull \$219.48

Arena. Multi-cancer early detection (MCED) blood screening — GRAIL (Galleri, 50+ cancers, only RCT) vs Exact Sciences (Cancerguard + Cologuard cash engine), Guardant (Shield, FDA-approved CRC → MCED), Freenome, Delfi; the race to originate the first FDA-approved, reimbursed MCED franchise.

POWER ORIENTATION
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	A reimbursed-volume cost curve is future; today's self-pay volume is sub-scale.
Network Economies	No network flywheel yet; data scale is the closest analog.
Counter-Positioning	Galleri's 50-cancer breadth + build-own dataset is hard for single-cancer (Shield CRC) entrants to mirror without their own large trials.
Switching Costs	Provider-ordering and EHR integration build mild lock-in; early.
Branding	The MCED category-definer brand; strong but dented by the NHS miss.
Cornered Resource · thesis leans here	The only randomized MCED dataset (NHS ~140k) + first PMA + largest commercial test base — the strongest, if contested, origination asset.
Process Power	Assay + bioinformatics pipeline is real but not a demonstrated durable process edge.

THE RACE · NAMED RIVALS

Exact Sciences	~\$3B-rev co.; self-funded
~25% terminal share · Cancerguard MCED (LDT, launched 2025) funded by the profitable Cologuard franchise; huge commercial muscle.	
Guardant Health	~\$700M-rev co.
~22% terminal share · Shield FDA-approved for CRC (Medicare \$1,495); extending to MCED (Breakthrough Device).	
Freenome	\$1.35B private
~10% terminal share · CRC-first → multi-cancer; ~\$1.35B raised (Roche, T. Rowe).	
Delfi Diagnostics	Merck GHI + rounds
~5% terminal share · FirstLook Lung (fragmentomics); niche, lung-first.	

LEAD / LAG · FALSIFIERS

MCED clinical evidence (RCT)	NHS ~140k (only RCT; primary missed) vs none randomized	LEADING
FDA approval	PMA filed (decision 2026-27) vs Guardant Shield approved (CRC)	ROUGHLY LEVEL
Reimbursement / commercial scale	self-pay; Medicare gated to 2028 vs Exact Cologuard reimbursed engine	LAGGING

COMPETITIVE READ

GRAIL is originating the MCED category — the broadest test (50+ cancers), the only randomized dataset, and the first PMA — but the NHS-Galleri primary-endpoint miss dented the 'proven population benefit' narrative, and Exact (Cologuard cash engine) and Guardant (FDA + Medicare in CRC) are scaling MCED behind it. The origination window is open but closing: GRAIL must convert the PMA into approval + Medicare coverage before better-capitalized rivals scale, and the extended NHS mortality follow-up is the falsifier that could reopen — or close — the whole thesis.

GRAIL show your work

Bear \$12.82 Base \$73.13 Bull \$219.48 · Weighted \$79.54 (+9.0%)

ULTRA BEAR

\$1.69

ASSUMPTIONS				
TAM Year-10 (\$B)	25			
Mature share (%)	1.3			
Mature op margin (%)	3			
Sales-to-capital	1.0			
Terminal WACC (%)	11.5			
Terminal growth (%)	2.0			
P(failure) (%)	40			
Scenario probability (%)	20			

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.18	-140%	-0.28	-0.25
FY28	0.22	-110%	-0.28	-0.21
FY29	0.25	-80%	-0.23	-0.15
FY30	0.27	-55%	-0.17	-0.10
FY31	0.29	-35%	-0.12	-0.06
FY32	0.30	-22%	-0.08	-0.04
FY33	0.31	-12%	-0.05	-0.02
FY34	0.31	-5%	-0.01	-0.01
FY35	0.32	+0%	-0.01	-0.00
FY36	0.32	+3%	+0.01	+0.00

Σ PV explicit FCF	-0.84
+ PV terminal (g 2.0%)	0.03

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-0.81B
+ Cash & investments	\$0.82B
= Equity value	\$0.01B
Raised over period	\$0.70B
Dilution by FY36	+50%
FY36 shares (M)	66

DCF per share	\$0.15
× (1-P_fail) [60%]	\$0.09
+ P_fail × distress [40% × \$4.00]	\$1.60

= Expected per share \$1.69

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$2.91	\$5.02	\$8.65	\$14.91
0.04×	0.07×	0.12×	0.20×

BEAR

\$12.82

ASSUMPTIONS				
TAM Year-10 (\$B)	25			
Mature share (%)	4.1			
Mature op margin (%)	18			
Sales-to-capital	1.1			
Terminal WACC (%)	10.0			
Terminal growth (%)	2.5			
P(failure) (%)	22			
Scenario probability (%)	28			

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.19	-120%	-0.27	-0.23
FY28	0.26	-80%	-0.27	-0.21
FY29	0.36	-45%	-0.25	-0.17
FY30	0.50	-20%	-0.23	-0.14
FY31	0.65	+0%	-0.14	-0.07
FY32	0.78	+8%	-0.06	-0.03
FY33	0.88	+13%	+0.02	+0.01
FY34	0.95	+16%	+0.09	+0.03
FY35	1.00	+18%	+0.14	+0.05
FY36	1.03	+18%	+0.16	+0.05

Σ PV explicit FCF	-0.71
+ PV terminal (g 2.5%)	0.70

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-0.01B
+ Cash & investments	\$0.82B
= Equity value	\$0.81B
Raised over period	\$0.65B
Dilution by FY36	+27%
FY36 shares (M)	56

DCF per share	\$14.46
× (1-P_fail) [78%]	\$11.28
+ P_fail × distress [22% × \$7.00]	\$1.54

= Expected per share \$12.82

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$20.65	\$33.25	\$53.55	\$86.25
0.28×	0.46×	0.73×	1.18×

BASE

\$73.13

ASSUMPTIONS				
TAM Year-10 (\$B)	25			
Mature share (%)	10.4			
Mature op margin (%)	30			
Sales-to-capital	1.3			
Terminal WACC (%)	10.0			
Terminal growth (%)	2.5			
P(failure) (%)	12			
Scenario probability (%)	34			

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.20	-100%	-0.24	-0.21
FY28	0.30	-50%	-0.23	-0.18
FY29	0.48	-15%	-0.21	-0.15
FY30	0.75	+5%	-0.17	-0.11
FY31	1.15	+15%	-0.14	-0.07
FY32	1.60	+22%	+0.01	+0.00
FY33	2.00	+26%	+0.21	+0.10
FY34	2.30	+29%	+0.44	+0.18
FY35	2.50	+30%	+0.60	+0.22
FY36	2.60	+30%	+0.70	+0.24

Σ PV explicit FCF	+0.02
+ PV terminal (g 2.5%)	3.24

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$3.26B
+ Cash & investments	\$0.82B
= Equity value	\$4.08B
Raised over period	\$0.40B
Dilution by FY36	+14%
FY36 shares (M)	50

DCF per share	\$81.60
× (1-P_fail) [88%]	\$71.81
+ P_fail × distress [12% × \$11.00]	\$1.32

= Expected per share \$73.13

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$117.78	\$189.68	\$305.48	\$491.98
1.61×	2.60×	4.18×	6.74×

BULL

\$219.48

ASSUMPTIONS				
TAM Year-10 (\$B)	25			
Mature share (%)	18.0			
Mature op margin (%)	36			
Sales-to-capital	1.5			
Terminal WACC (%)	9.0			
Terminal growth (%)	3.0			
P(failure) (%)	6			
Scenario probability (%)	13			

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.21	-80%	-0.21	-0.19
FY28	0.35	-30%	-0.20	-0.16
FY29	0.60	+5%	-0.14	-0.10
FY30	1.00	+18%	-0.09	-0.06
FY31	1.70	+26%	-0.03	-0.01
FY32	2.50	+31%	+0.24	+0.13
FY33	3.20	+34%	+0.62	+0.29
FY34	3.80	+35%	+0.93	+0.40
FY35	4.20	+36%	+1.25	+0.49
FY36	4.50	+36%	+1.42	+0.51

Σ PV explicit FCF	+1.31
+ PV terminal (g 3.0%)	8.79

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$10.10B
+ Cash & investments	\$0.82B
= Equity value	\$10.92B
Raised over period	\$0.20B
Dilution by FY36	+7%
FY36 shares (M)	47

DCF per share	\$232.34
× (1-P_fail) [94%]	\$218.40
+ P_fail × distress [6% × \$18.00]	\$1.08

= Expected per share \$219.48

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$337.70	\$519.59	\$799.45	\$1,230.06
4.63×	7.12×	10.95×	16.85×

ULTRA BULL

\$444.35

ASSUMPTIONS				
TAM Year-10 (\$B)	25			
Mature share (%)	27.2			
Mature op margin (%)	40			
Sales-to-capital	1.7			
Terminal WACC (%)	8.5			
Terminal growth (%)	3.0			
P(failure) (%)	3			
Scenario probability (%)	5			

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.22	-60%	-0.17	-0.16
FY28	0.40	-10%	-0.15	-0.12
FY29	0.72	+15%	-0.08	-0.06
FY30	1.30	+27%	+0.01	+0.01
FY31	2.20	+33%	+0.20	+0.12
FY32	3.30	+37%	+0.57	+0.31
FY33	4.40	+39%	+1.07	+0.53
FY34	5.40	+40%	+1.57	+0.72
FY35	6.20	+40%	+2.01	+0.84
FY36	6.80	+40%	+2.37	+0.91

Σ PV explicit FCF	+3.11
+ PV terminal (g 3.0%)	17.10

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$20.21B
+ Cash & investments	\$0.82B
= Equity value	\$21.03B
Raised over period	\$0.20B
Dilution by FY36	+5%
FY36 shares (M)	46

DCF per share	\$457.17
× (1-P_fail) [97%]	\$443.45
+ P_fail × distress [3% × \$30.00]	\$0.90

= Expected per share \$444.35

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$668.15	\$1,004.67	\$1,510.68	\$2,271.54
9.15×	13.76×	20.69×	31.12×

GRAIL supporting analysis · disclaimers · glossary

Bear \$12.82 Base \$73.13 Bull \$219.48

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

Only randomized MCED dataset.
The NHS-Galleri RCT (~140k) is the largest, only randomized MCED dataset — a cornered resource even after the primary-endpoint miss (stage IV fell -14%; mortality follow-up extended).
- 2

FDA filing rests on PATHFINDER 2, not NHS.
The PMA is built on the positive PATHFINDER 2 (69.8% sensitivity for the 12 deadliest, 99.6% specificity) plus the NHS prevalent round — not the failed late-stage primary.
- 3

Medicare law already signed.
The MCED coverage benefit is law (Feb 2026); coverage is gated on FDA approval and earliest 2028, but the legislative risk is retired.
- 4

Cash-funded into the coverage window.
~\$823M and burn halved to ~\$300M/yr funds GRAIL into the ~2028 Medicare window without an imminent raise.
- 5

Tiny share count amplifies the unlock.
~43M shares means a reimbursed-volume inflection moves per-share value hard — the asymmetry cuts both ways.

FALSIFICATION TRIGGERS

Bull validation FDA PMA approval (2026-27) · Medicare NCD activates (~2028) · broad private coverage · test volume sustains >40% growth	Bear validation PMA rejected/delayed · Medicare coverage narrow or post-2028 · private payers demand mortality data	Reframe needed if the extended NHS follow-up shows a mortality benefit, the clinical-utility debate resolves and the bull/ultra weights rise materially
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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Galleri — GRAIL's blood test screening for 50+ cancers from one draw; sold self-pay (~\$949) today as an LDT, FDA PMA filed Jan 2026.	MCED — Multi-cancer early detection — blood-based screening across many cancer types; a new category GRAIL pioneered.	NHS-Galleri — The ~140k-participant UK randomized trial; missed its stage-shift primary endpoint (Feb 2026) but cut stage IV; mortality follow-up extended.
PMA / NCD — FDA Premarket Approval (filed Jan 2026) and Medicare National Coverage Determination (earliest 2028) — the two gates to reimbursed scale.	Sales-to-capital (s2c) — Damodaran reinvestment efficiency — incremental revenue per dollar of reinvested capital.	